MIL-R-38953C(USAF)
25 April 1973

SUPERSEDING
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3 June 1968

MILITARY SPECIFICATION

RACK, BOMB EJECTOR, AIRCRAFT MAU-12B/A AND RACK, BOMB EJECTOR, AIRCRAFT MAU-12C/A

*1. SCOPE

1.1 Scope. This specification covers the requirements for one type of equipment designated as Rack, Bomb Ejector, Aircraft MAU-12B/A and Rack, Bomb Ejector, Aircraft MAU-12C/A, hereafter referred to as the rack for carrying and either freefall release or ejection of stores weighing up to 5000 pounds.

2. APPLICABLE DOCUMENTS

*2.1 The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-C-843	Cushioning Material, Cellulosic

Military

MIL-P-116	Preservation, Methods of
MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-T-7743	Testing, Store Suspension Equipment, General Specification For
MIL-Q-9858	Quality Control System Requirements
MIL-C-45662	Calibration System Requirements

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence For The Selection Of
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

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MIL-STD-785	Requirements for Reliability Program (For Systems and Equipment)
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment
MIL-STD-1186	Cushioning, Anchoring, Bracing Blocking, and Waterproofing; With Appropriate Test Methods

DRAWINGS

Air Force

64J13210	Rack Assembly, Bomb Ejector, Aircraft MAU-12B/A
64D13204	Diagram, Schematic, Electrical
64J13010	Interconnecting Group KMU-237/B
64D13265	Retainer, Cartridge
69J13060	Rack Assembly, Bomb Ejector, Aircraft MAU-12C/A

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

*3.1 Qualification. The racks furnished under this specification shall be products that have been tested and passed the qualification tests specified herein, and have been listed or approved for listing on the applicable qualified products list at the time set for opening of bids.

*3.2 Components

*3.2.1 Selection of materials. Specifications and standards of all materials and parts, and Government certification and approval of processes and equipment which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143.

*3.2.2 The MAU-12B/A and MAU-12C/A Racks shall comprise an integral assembly consisting of:

a. Cartridge actuated release mechanism which will accommodate stores having 14 inch or 30 inch lug spacing and diameters from 10.70 inches to 33 inches.

b. Solenoid actuated in-flight locking mechanism to prevent inadvertent release of special stores.

c. Controlled store ejection system with two ejection pistons which automatically return to their retracted positions at the completion of their ejection strokes.

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*3.3 Design and construction. The racks shall be constructed in conformance to drawings 64J13210 (MAU-12B/A) or 69J13060 (MAU-12C/A) and subsidiary drawings.

3.3.1 Over center hook linkage. Each rack shall be equipped with an ajustable over center lock release system.

3.3.2 Interchangeability. The rack shall be completely interchangeable from airplane to airplane. Parts functionally and dimensionally interchangeable shall be identified by the same part number. When parts are not interchangeable, the criteria listed in MIL-D-1000 shall apply and new drawings shall be prepared as required.

*3.4 Performance. The rack shall be capable of releasing stores by means of a cartridge-actuated mechanism which shall allow either freefall drop of stores or forced ejection of stores of all types weighing up to and including the 5000 pound class stores, having either 14 inch or 30 inch lug spacing and having a minimum diameter of 10.70 inches and a maximum diameter of 33 inches.

3.4.1 Forced ejection. When forced ejection of the store is desired, the velocity of store, reaction forces and the flight attitude of the store shall be controlled by orifices which are placed in the gas system and which meter the flow of gases to the ejection pistons.

3.4.2 Locked shut and sympathetic firing. The rack shall be capable of withstanding the stresses incurred when fired with the inflight safety lock engaged without inadvertently releasing the store or damage to any structural part of the rack. Both cartridges shall fire when only one of the firing switches (S-1, S-3) is closed.

3.4.3 In-flight lock. With the in-flight lock in the locked position, no electrical energy shall be transmitted to the cartridge electrical terminals when a 28 volt potential is applied to the cartridge firing circuits.

3.4.4 Provisions for conventional weapons

*3.4.4.1 Arming controls. (Conventional weapons). The nose, center, and tail solenoid operated arming controls shall operate from an 18 to 30 vdc power supply and shall each draw a maximum of 0.15 amperes continuously when energized at 24 vdc. Each control shall be capable of receiving and holding all standard arming wire loops. Each control shall be selectively capable of providing for an unarmed release or ejection, or armed release or ejection, of stores weighing from 100 pounds through the 5000 pound class.

3.4.4.2 Fuzing provision. When electrical fuzing of conventional weapons is required, the Interconnecting Group KMU-237/B shall be incorporated in the rack in conformance to drawing 64J13010.

3.4.5 Life. The rack shall function satisfactorily for 250 firings.

3.4.6 Reliability. The contractor shall initiate and maintain a reliability program in accordance with MIL-STD-785. This program shall be subject to modification by the procuring activity.

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3.4.6.1 Reliability levels. The following reliability levels shall be achieved and demonstrated by the contractor:

a. Inadvertent release. With proper adjustment of the over center linkage, there shall be no inadvertent release during the design service life of the rack (250 firings under service conditions).

b. Release performance requirements. With proper maintenance and replacement of parts, the achieved rate of failure shall be no failures in 250 releases under service conditions, excluding cartridge failure.

c. Reliability failure. The bomb rack shall be constructed as specified herein and successfully complete the tests of paragraph 4.6 of this specification with no failures as defined herein preventing satisfactory performance. Reliability of 98.0 per cent for a 2-hour estimated mission time with Mean Time Between Failure (MTBF) value of 100 hours is required. Compliance will be acceptably demonstrated at 90 per cent confidence level by 440 hours or more of total qualification testing time with no uncorrected failures.

3.5 Environmental requirements. The rack shall be capable of proper operation during or after, as applicable, in the following environmental conditions:

* a. Temperature from -54°C (-65°F) to 93°C (200°F) for the MAU-12B/A Rack and -54°C (-65°F) to 176.7°C (350°F) for the MAU-12C/A Rack.

b. Humidity up to 95 per cent, relative.

c. Salt spray incident to use, storage or transportation near or on a large body of salt water.

d. Icing conditions incident to service use.

e. Sand and dust likely to be enountered under desert conditions.

f. Vibration incident to flight, transportation and handling.

g. Shock incident to flight, transportation and handling.

3.6 Identification of product. The rack and all component parts thereof shall be marked for identification in accordance with MIL-STD-130.

*3.7 Government furnished property. When specified in the contract, the Government will furnish the following to the contractor, upon his request, for use in testing the racks, Bomb Ejector, Aircraft MAU-12B/A and MAU-12C/A:

	Item Number	Part Number	Description of Item	Quantity
*	1	P7877-1	Cartridge, Bomb Ejection, ARD 446-1	As stated in the contract
*	2	10-3400	Cartridge, Bomb Ejection, ARD 863-1	As stated in the Contract
*	3	67F14000	Cartridge, Bomb Ejection, CCU-1/B	As stated in the Contract

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3.8 Workmanship. The rack shall be fabricated and finished in a manner which will insure that the criteria of appearance, fit, smoothness of operation, and adherence to specified tolerances are observed. These criteria must apply to all accessories and to all external non-functioning parts, such as nameplates, as well as to functional parts. Particular attention shall be given to neatness and thoroughness of marking of parts and assemblies, plating, painting, riveting, machine screw assemblage, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the qualifying activity. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the rack shall be as follows:

a. Qualification tests

b. Acceptance tests

4.3 Qualification testing.

*4.3.1 Testing. Qualification testing of preproduction rack(s) shall be performed by an independent laboratory acceptable to the activity responsible for the Qualified Products List (see paragraph 6.5). Racks shall have been produced with manufacturing equipment and procedures normally used in production.

*4.3.2 Qualification test reports. Qualification data shall be prepared and submitted for evaluation in accordance with the requirements of the procuring activity.

*4.3.3 Qualification test requirements. Qualification tests shall consist of all tests specified herein under "4.6 Test methods" as listed in Table I.

4.4 Acceptance tests. Acceptance testing shall consist of the following:

a. Individual tests

b. Sampling tests

4.4.1 Individual tests. Each rack shall be subjected to the individual tests as listed in Table I, except as otherwise specified.

4.4.2 Sampling plan and tests.

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*4.4.2.1 Sampling plan. One sample shall be selected at random by the quality assurance representative from each 200 production racks. Prior to being subjected to the sampling tests, the sample racks shall have passed the individual tests.

4.4.2.2 Sampling tests. Sampling tests shall consist of the sampling tests shown in Table I.

4.4.3 Rejection and retest. When the sample rack fails to meet the specified test, no rack on hand or later produced shall be accepted until the extent and cause of failure are determined. The failure shall be corrected and the unit shall demonstrate compliance by satisfactorily passing the failed test before acceptance of on hand or later produced units.

4.4.4 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.5 Test conditions. Test conditions shall be as specified in each specific test under "4.6 Test methods."

TABLE I

TEST	REQUIREMENT PARAGRAPH	TEST PARAGRAPH	QUALIFICATION TEST	INDIVIDUAL TEST	SAMPLING TEST
Examination of Product	3.3, 3.6, 3.8	4.6.1	X	X	
Conformance to Drawings	3.3, 3.8	4.6.2	X	X	
Static Load Tests	3.3	4.6.3	X		X
*Functional Test	3.4	4.6.4	X	X	
**Locked Shut and Sympathetic Firing Test	3.4	4.6.5	X	X	
Reliability Life Test	3.4	4.6.6	X		X
Environmental Tests	3.5	4.6.7			
High Temperature	3.5.a	4.6.7.1, 4.6.7.1.1	X X		X X
Humidity Test	3.5.b	4.6.7.2	X		
Salt Fog Test	3.5.c	4.6.7.3	X		X
Icing Conditions Test	3.5.d	4.6.7.4	X		X

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TABLE I (Cont'd)

TEST	REQUIREMENT PARAGRAPH	TEST PARAGRAPH	QUALIFICATION TEST	INDIVIDUAL TEST	SAMPLING TEST
Sand and Dust Test	3.5.e	4.6.7.5	X		
Vibration Test	3.5.f	4.6.7.6	X		X
Shock Test	3.5.g	4.6.7.7	X		
Electrical Tests	3.2	4.6.8	X	X	X

*These tests may be performed concurrently with the Reliability Life Test in Qualification Testing.

**Every twentieth rack shall be tested to this test (individual tests only).

4.6 Test methods.

4.6.1 Examination of product. Prior to installation of the left side plate each rack shall be examined visually to determine compliance with all requirements of this specification for which tests are not specified. Failure to comply with all requirements shall be cause for rejection.

4.6.2 Conformance to drawings. Each rack shall be inspected to determine conformance to the engineering drawings, subsidiary drawings and specifications and standards referenced therein. Failure to conform to these criteria shall be cause for rejection.

4.6.3 Static load tests.

NOTE: Tolerances for all loads applied in the static load tests shall be ± 3 per cent.

4.6.3.1 Yield load tests.

4.6.3.1.1 Thirty-inch suspension.

*4.6.3.1.1.1 Test set up. The rack shall be installed in a fixture capable of withstanding the loads specified in the test procedures. This fixture shall be rigidly constructed in a manner such that vertical deflection of the bomb rack shall not exceed 1/2 inch during performance of the firing tests specified in the functional test. Pins or bolts for this installation shall have a minimum ultimate tensile strength of 160,000 psi. A 2000 pound simulated store as shown in Figure 2 shall be installed on the 30-inch hooks of the rack. The set of sway braces at each end of the rack shall be torqued to 840 ± 50 inch-pounds. After the studs are in contact with the store, the torque shall be applied by turning each bolt of a set the same number of turns ± 1/8 turn. A recommended fixture is shown in Figure 1.

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4.6.3.1.1.2 Test procedure. The yield load test of the rack shall consist of the simultaneous imposition of loads for each condition along or about, as applicable, the axes shown in Figure 2. For each condition the loads shall first be imposed in increments of 10 per cent of design load until 100 per cent of design load is reached, then in increments of 5 per cent of design load until 115 per cent of design load is reached. The loads shall be held at each per cent increment for three minutes before proceeding to the next per cent increment after 60 per cent of the design load is attained. The rate of loading shall not exceed 5 per cent of the design loads per second. When a load of 115 per cent of design load has been reached and held for three minutes, the load shall be relieved to zero and the rack inspected. The simulated store shall be released and the position of the rack in the fixture shall be reversed. Test set up and procedures shall be repeated. Any permanent deformation shall be cause for rejection. Design loads shall be as shown in Table II.

TABLE II

DESIGN LOADS

30 INCH HOOKS (MAU-12B/A						
CONDITION NUMBER	P_x Lbs	P_y Lbs	P_z Lbs	M_x In-Lbs	M_y In-Lbs	M_z In-Lbs
1	887	2240	-11933	4473	-88867	302533
2	2000	1633	-12600	0	-214000	231333
3	2000	17933	-12600	0	-256800	213600
4	193	4687	-10800	4300	-21133	-301200

30 INCH HOOKS (MAU-12C/A						
CONDITION NUMBER	P_x Lbs	P_y Lbs	P_z Lbs	M_x In-Lbs	M_y In-Lbs	M_z In-Lbs
1	3050	15600	-12300	36000	292000	-165000
2	3050	10800	-20300	36000	384000	-110000

Reference axes and positive directions of those axes are shown in Figure 2.

4.6.3.1.2 Fourteen inch suspension.

4.6.3.1.2.1 Test set up. Test set up shall be as specified for the 30 inch suspension except that the simulated store shall be suspended on the 14 inch hooks of the rack as shown in Figure 3.

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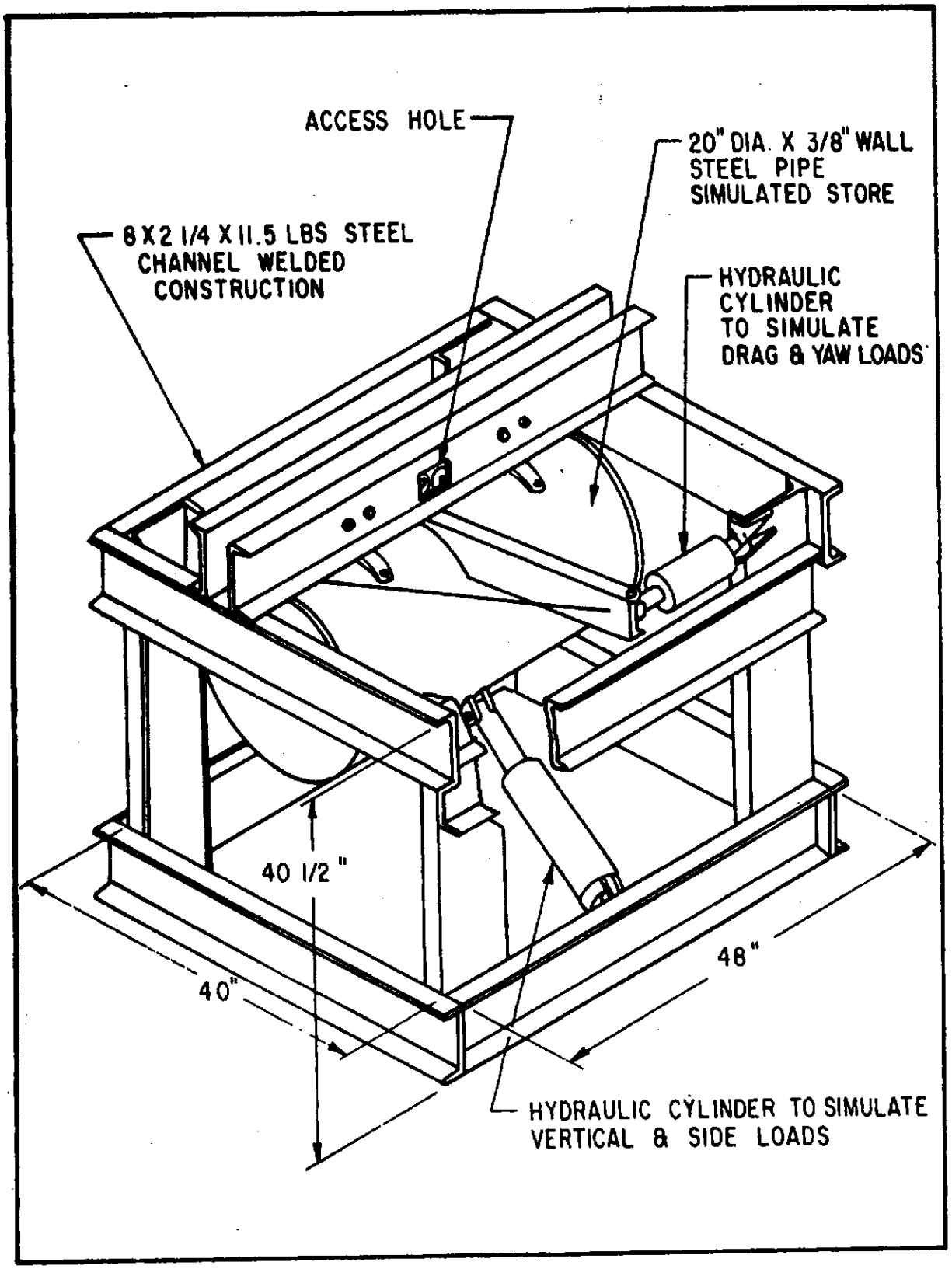


FIGURE 1 STATIC LOAD FIXTURE

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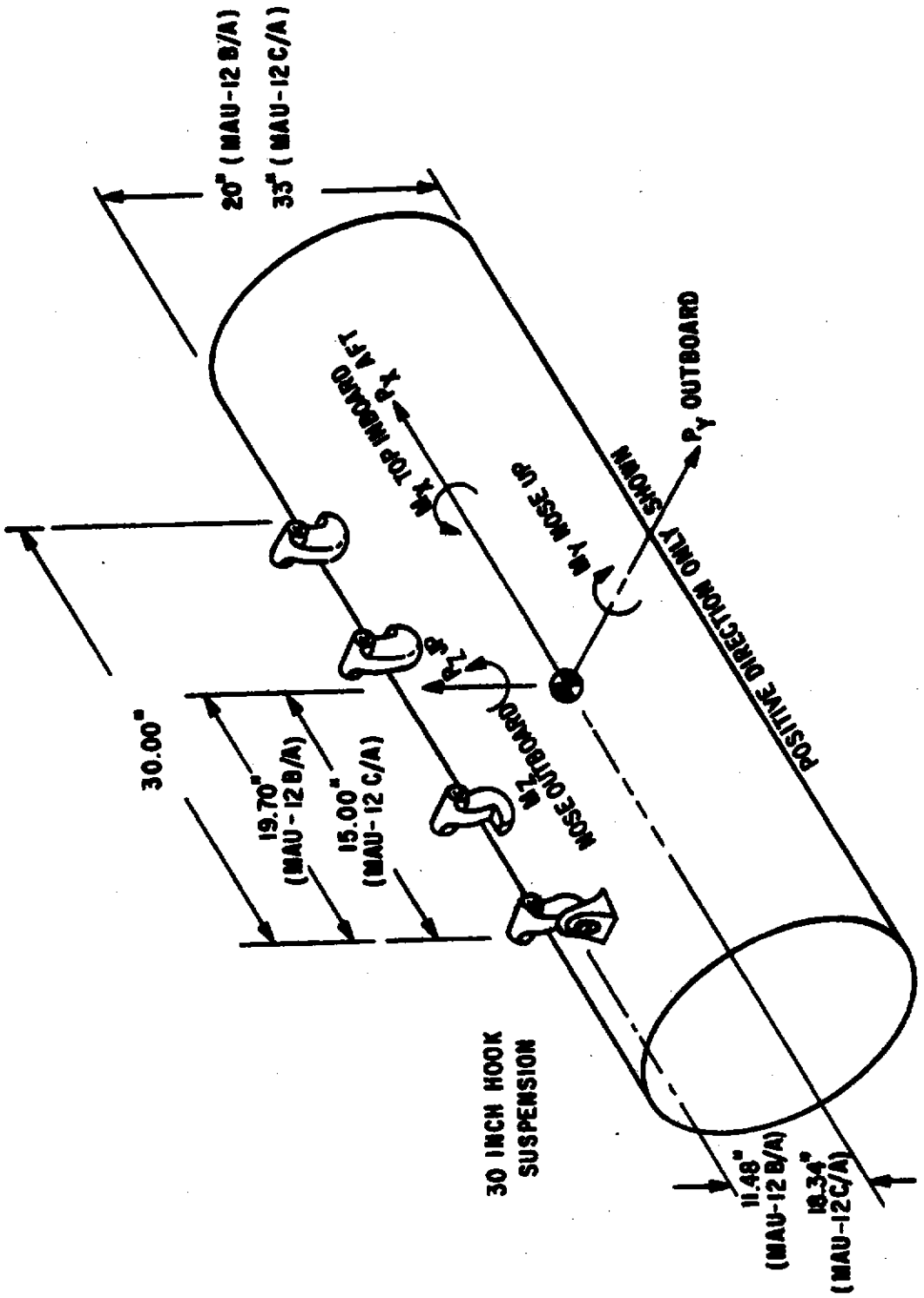


FIGURE 2 LOCATION AND DIRECTION OF LOADS

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4.6.3.1.2.2 Test procedure. The test procedure shall be as specified for the thirty inch suspension. Design loads shall be as shown in Table III. The rejection criteria for the 30 inch suspension also applies to the 14 inch suspension.

TABLE III

14 INCH HOOKS						
CONDITION NUMBER	P_x lbs	P_y lbs	P_z lbs	M_x in-lbs	M_y in-lbs	M_z in-lbs
1	350	5610	-3220	10000	-28400	45280
2	240	6740	-3840	15800	-23000	31340
3	350	6975	-4170	16800	-21700	31000
4	670	2935	-4000	0	-125300	83500

4.6.3.2 Ultimate strength test.

4.6.3.2.1 Thirty inch suspension.

*4.6.3.2.1.1 Test set up. The test set up for this test shall be as specified for the YIELD LOAD TEST.

4.6.3.2.1.2 Test procedure. Test procedure shall be as specified for the YIELD LOAD TEST up to 115 per cent of the design load, except that after 115 per cent of the design load has been reached, loading shall continue in 5 per cent increments of the design load up to 150 per cent of the design load. After 150 per cent of the design load has been reached and held for three minutes, the load shall be reduced to zero and the rack inspected. The rack shall be tested in one direction only. Any breakage during loading up to and including 150 per cent of design loads shall be cause for rejection. Design loads shall be as shown in condition 3 only of Table II for the MAU-12B/A and in condition 2 only of Table II for the MAU-12C/A.

4.6.4 Functional test. Testing of the rack shall be performed as outlined in the following procedures and in the order listed:

a. The rack shall be mounted on a test stand in a horizontal position with the hooks downward.

b. A 2000 pound simulated store with standard Air Force bomb lugs on 30 inch centers and the center of gravity located midway between the lugs shall be suspended from the rack. The ground safety pin shall be properly inserted. Sway braces shall be torqued as required in the test set up of the 30 inch suspension yield load test paragraph of this specification. The ground safety pin shall be removed and reinserted. Any movement of the hook linkage or

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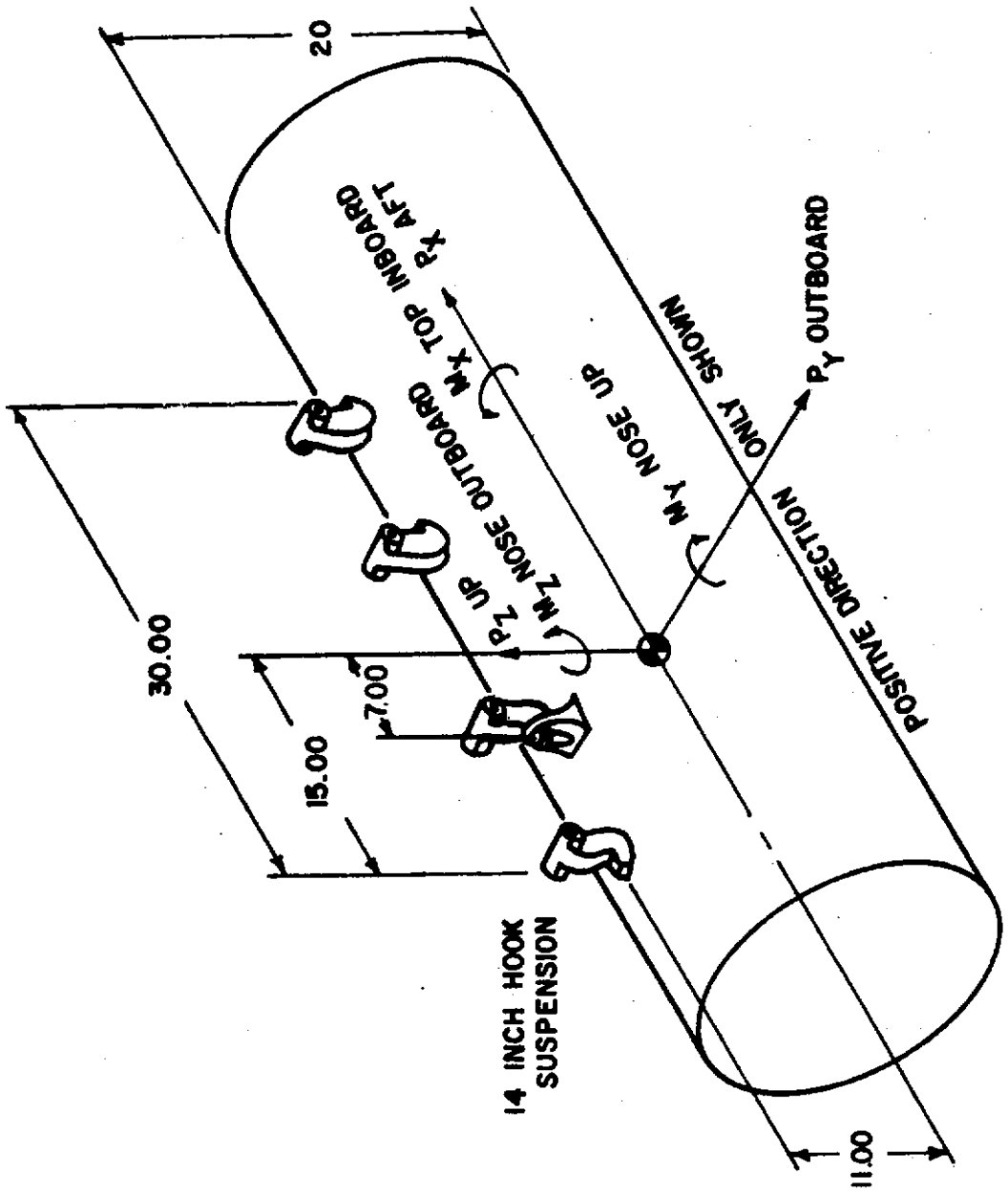


FIGURE 3 LOCATION AND DIRECTION OF LOADS

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rotation of the center bellcrank that prevents or restricts the removal or re-insertion of the ground safety pin shall be cause for rejection. Two ARD 446-1 cartridges shall be inserted in the cartridge retainers (P/N 64D-13265) and installed in the breech of the rack. The retainers shall be screwed into the breech with a torque of 250 to 300 inch pounds. Forward and aft orifice pin assemblies with a diameter of 0.111 inches shall be installed. The inflight safety lock shall be unlocked by applying a 28 ± 4 volt current across pins L and M of the electrical connector. With the ground safety pin removed, 28 ± 4 volts of DC shall be applied to Pin A-j and C-j and the cartridges shall fire. The hooks shall open and the store shall be forcibly ejected. The velocity of the store at the end of the piston stroke shall be 14 ± 2 feet per second. The ejector pistons shall have returned to their retracted positions. In all cases except individual tests, the peak reaction forces against the supporting structure shall be measured and recorded for each piston. The flight attitude of the store at a position between 2.5 and 3.0 feet below the suspended position shall be horizontal ± 1-1/2 degrees.

* c. The rack hooks shall be latched by clockwise rotation of a suitable tool engaged in the square manual release socket located on the side of the rack containing the cartridge access to the breech and designated by the "Hook Manual Release" presentation. Torque required for this operation shall be greater than 35 inch pounds, but not more than 80 inch pounds. At this time the inflight safety lock arm shall be checked through the inspection window in the side of the rack to determine that it is in the fully locked position. This is shown as the "LOCK" position on each side of the rack. If the inflight safety lock does not self-rotate from spring loading to the fully locked position, the rack shall be rejected.

d. With the rack latched and the inflight safety lock arm in locked position as indicated in step c. of this procedure, the manual release tool shall be inserted in the manual release socket and rotation attempted in the direction indicated for manual release. The hooks shall not open.

* e. With the rack latched, the inflight safety lock shall be unlocked by inserting a suitable tool in the hexagonal socket located on the unlocked left side of the rack and rotated until the inflight safety lock arm is in the unlocked position as shown by the "LOCK-UNLOCK" presentation on the side of the rack. Torque required for this operation shall be 20 inch pounds maximum. This operation is shown by the "Inflight Lock Manual Release" presentation on the side of the rack. The safety lock bellcrank shall be pinned to the unlocked position by insertion of 0.248 inch maximum, 0.245 inch minimum, diameter pin through the sideplates and the safety lock switch actuating arm. The hook manual release shall be rotated in a counterclockwise direction. Torque required for this operation shall be 45 ± 15 inch pounds. The hooks shall open and remain in an opened position. All circuits of switches S-1 and S-3 shall be checked for proper continuity in accordance with drawing 64D13204. NOTE: The torque measurements of steps c., d., and e., shall be repeated three times.

f. The rack hooks shall be latched and 28 ± 4 volts DC applied to pins (L) and (M) of the electrical plug located at the end of the rack. The inflight safety lock shall open and remain open. Maximum solenoid actuation current shall be 15.5 amps. Maximum holding current shall be 0.175 amps. The hooks, shall be manually opened, and remain in an opened position.

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g. The store away switch shall actuate when the end of the plunger is at a point 1.50 inches minimum below the suspension surface of the 14 inch hooks. The store away switch shall be monitored for proper actuation. Pins S-T, V-W, a-b, d-e shall indicate "store off" and pins T-U, W-X, b-c, e-f shall indicate "store present."

h. A standard arming wire for conventional weapons shall be installed in each of the arming controls. Using a convenient device, the force required to pull the arming wires installed in the arming solenoid shall be measured. This measurement shall be made twice for each arming control, once with the arming solenoid energized. The required forces shall be:

(1) Arming solenoid not energized:

The arming control shall not release the arming wire when a force of 10 pounds is applied. The arming control shall release the arming wire when a force of 14 pounds is applied.

* (2) Arming solenoid energized:

The arming control shall not release the arming wire when a force of 150 pounds is applied.

NOTE: Throughout this specification, when orifice diameters are specified, the diameter tolerances shall be those shown in the specific orifice drawing.

4.6.5 Locked shut and sympathetic firing test. The rack shall be mounted with the top closeout channel removed. The pin connecting S-1 to the safety lock linkage shall be removed through the appropriate hole in the left side plate. A 2000 pound simulated store shall be loaded on the rack. Two ARD 446-1 shall be installed and the inflight safety lock engaged. Special orifice assemblies shall be installed with a 0.025 minimum to 0.040 maximum inch diameter. The safety lock bellcrank shall remain in the spring-loaded, locked position. S-1 shall be actuated by manually closing its firing circuit. The rack shall be fired. Both cartridges shall have fired. The rack hooks shall not unlatch and the store shall remain suspended on the rack. After the Locked Shut and Sympathetic Firing Test, the rack shall be subjected to steps c., d., and e., of the Functional Test. Failure to meet these criteria or evidence of failure or permanent deformation of any structural part shall be cause for rejection.

*4.6.6 Reliability life test. Perform step b. of the functional test. After completion of step b., of the functional test, 100 releases of the 2000 pound simulated store shall be performed at ambient temperature with the fore and aft orifice pin assemblies having 0.111 inch diameters and 100 releases of the 500 pound simulated store with the fore and aft orifice pin assemblies having 0.093 inch diameters. Two ARD-446-1 cartridges or CCU-1/B cartridges shall be used in each of the above firings. Using one ARD-446-1 or CCU-1/B and one ARD-863-1 cartridge, or equal, 50 releases of a 100 pound simulated store shall be performed with the fore and aft orifice pin assemblies having 0.156 inch diameters. The two ejector pistons and the slave piston shall be removed from the rack, disassembled and cleaned after each 25 firings or at the end of

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each firing day, whichever occurs first. The two ejector pistons shall retract completely for the first five firings following each cleaning of the rack. After each fiftieth firing, orifice pin assemblies, and all "O" rings in the slave and ejector piston assemblies shall be replaced. Completion of this test without such damage or erosion which precludes satisfaction of the functional test shall be considered as meeting the operational life requirements of this specification. Evidence of such damage, wear or erosion shall be cause for rejection.

*4.6.7 Environmental tests. During or after, as applicable, each environmental test, the rack shall be subjected to steps c., d., e., f., g., and h., of the functional test unless otherwise specified herein. Failure to meet the requirements of these portions of the functional test shall be considered as failure to pass the environmental tests and cause for rejection.

*4.6.7.1 High temperature. The rack shall be raised to and held at 176.7°C (350°F) for the MAU-12C/A and 93°C (200°F) for the MAU-12B/A, for four hours. The temperature shall be reduced to ambient and the rack shall be fired five times as specified in the firing portion of the functional test and inspection shall be made for gas leakage. In addition to the criteria in 4.6.7, evidence of gas leakage shall be cause for rejection.

4.6.7.1.1 The electrical assembly components only shall be raised to 176.7°C (350°F) and held for four hours. The temperature shall be reduced to ambient and the electrical assembly components shall meet the performance requirements of the high temperature test for the rack.

4.6.7.2 Humidity. The rack shall be subjected to the humidity test described in Procedure I, Method 507, MIL-STD-810.

4.6.7.3 Salt fog. The rack shall be subjected to a salt fog test in accordance with MIL-T-7743. Paragraph 4.12 of MIL-T-7743 shall apply.

*4.6.7.4 Icing conditions. With the hooks in the latched condition, the rack shall be stabilized at -54°C (-65°F), then subjected to 37.7°C (100°F) and 90 percent relative humidity until frost has disappeared. The rack shall then be stabilized at -54°C (-65°F). While the rack is at this temperature, step f. of the functional test shall be performed. Torque required for manual hook release shall not exceed 450 inch pounds.

4.6.7.5 Sand and dust. The rack shall be subjected to a sand and dust test in accordance with Procedure I, Method 510, MIL-STD-810.

*4.6.7.6 Vibration. The rack shall be subjected to a resonance search test and a resonance dwell test in accordance with MIL-STD-810, Method 514, except as specified herein. All electrical circuits shall be monitored during these tests. In addition to the criteria of 4.6.7, malfunction of any part or circuit, failure of any part, or inadvertent actuation of the store release mechanism during the vibration testing shall be cause for rejection.

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*4.6.7.6.1 Resonance search test. A frequency sweep shall be conducted in each test axis to determine the resonant frequencies of the rack/store combination. The location of the point(s) on the surface of the store which experiences the greatest vibration amplitude will be determined for each resonant frequency. The vibration of the store will be monitored and controlled at this point(s) by a vibration pickup(s) oriented to measure maximum store motion during resonance dwell test. The vibration amplitude during the frequency sweep should be the minimum required to provide the necessary data.

*4.6.7.6.2 Resonance dwell test. The rack shall be subjected to a resonance dwell test in accordance with MIL-STD-810, Method 514, Procedure I, Part I, except as specified herein. Resonance dwell tests will be conducted in each axis at each of the selected resonant frequencies. Figure 4 of this specification shall be utilized in lieu of the test curves of Figure 514.1 of MIL-STD-810. Testing in each axis shall be limited to the three resonant frequencies which exhibit the greatest vibration amplitude. The excitation frequency will be established at the appropriate resonant frequency. The vibration amplitude will then be increased until the maximum amplitude measured at the monitor location(s) equals the test curve value, (Figure 4) for that frequency. The resonance dwell will then be accomplished at this amplitude and frequency. In the event that resonant frequency shifts as amplitude increases, the input frequency will be established at the resonant frequency at test amplitude.

*4.6.7.6.2.1 Test conditions. Dwell time at each resonance will be thirty minutes. Fifteen minutes of dwell time will be made with the rack safety lock locked and the remaining 15 minutes with the safety lock unlocked. The complete three axis tests will be accomplished at both of the following conditions:

a. With 250 pound simulated store suspended from the 14 inch spacing shackles. The 250 pound store will have a minimum diameter of 10.75 inches.

b. With a 1000 pound simulated store suspended from the 30 inch spacing shackles.

*4.6.7.7 Shock. For this test, the maximum weight simulated store shall be the 2000 pound simulated store. Design loads shall be as illustrated in Table II. The rack shall be subjected to the yield load shock tests described in MIL-T-7743. This test shall be conducted with the safety lock locked and repeated with the safety lock pinned open.

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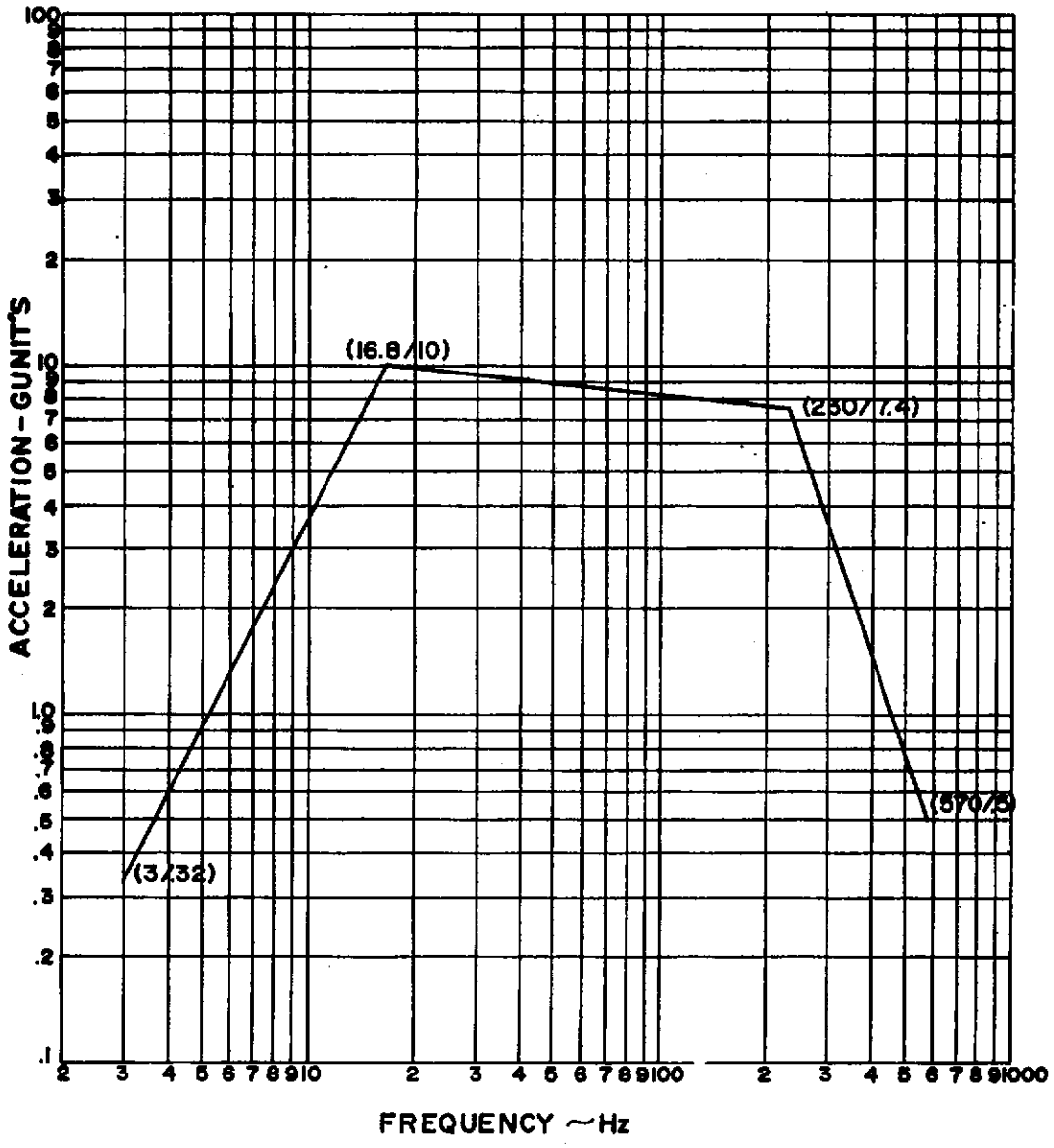


FIGURE-4 RESONANCE DWELL TEST CURVE

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*4.6.8 Electrical tests. After all preceding tests in 4.5 have been completed, a complete electrical continuity and insulation resistance test of all circuits as shown in Drawing 64D13204 shall be performed. The insulation resistance test shall be in accordance with MIL-STD-202, Method 302, Test Condition A. Insulation resistance shall be greater than 100 megohms. Failure to pass these tests shall be cause for rejection. Arming solenoids shall not be subjected to the insulation resistance test.

*4.7 Quality control requirements. The contractor shall provide and maintain a quality control system acceptable to the government for the equipment covered by this specification. The system of quality control shall be in accordance with specification MIL-Q-9858.

*4.7.1 Quality inspection equipment and facilities. Inspection equipment and facilities, shall be of sufficient accuracy, quality, and quantity to permit performance of the required inspection. The supplier shall establish and maintain a calibration system in accordance with MIL-C-45662.

5. PREPARATION FOR DELIVERY

*5.1 Levels A and B. Each rack shall be cleaned and dried in accordance with applicable processes and procedures of MIL-P-116, and individually preserved and packaged, one each, in accordance with Method IIB of MIL-P-116. All rack projections, sharp edges, and components shall be padded or wrapped with PPP-C-843 cushioning material. Inner container shall conform to the following specification: PPP-B-636, Type CF, Variety DW, Class Dom, Style RSC. Outer container shall conform to PPP-B-636, D.W., Style RSC, Class W/R, Grade V13C.

*5.1.1 Level C. Same as levels A and B, except barrier bag, desiccant and outer container shall be omitted.

5.2 Packing.

*5.2.1 Levels A and B. Racks preserved and packaged in accordance with 5.1 shall be overpacked in a PPP-B-601, Style A, overseas type shipping container in quantities which will not exceed weight limitation of specified container. Cushioning, anchoring, blocking, bracing and waterproofing shall be in accordance with MIL-STD-1186.

*5.2.2 Level C. Racks packaged in accordance with 5.1 and 5.1.1 which require overpacking for acceptance by the carrier shall be packed in an exterior type shipping container in a manner which will insure safe transportation at the lowest rate to the point of delivery. Containers shall meet consolidated freight classification rules or regulations of other common carriers as applicable to the mode of transportation.

*5.3 Marking for shipment and storage. Basic container markings (identification, contract data, address, shipping piece number, international logistic label, special markings, etc.), shall be in accordance with MIL-STD-129. The nomenclature shall be:

RACK, BOMB EJECTOR, AIRCRAFT MAU-12B/A or

RACK, BOMB EJECTOR, AIRCRAFT MAU-12C/A

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6. NOTES

*6.1 Intended use. The MAU-12B/A and the MAU-12C/A racks are universal racks designed to be attached to a variety of aircraft and will be used to carry and release special stores, conventional stores and fuel tanks.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Conditions under which Government loaned and Government furnished property will be made available to the contractor.

c. Required level of preservation, packaging and packing.

c. Number of spare "O" rings to be furnished with each rack.

6.3 Sample plan for vendors. In the event multiple vendors of a component are utilized, or a change in vendors occurs, the contractor is obligated to provide one additional representative bomb rack unit for sampling tests containing said multiple vendor part or parts changed from approved qualification sample supplier without an increase in price.

6.4 Definitions.

6.4.1 Acceptance tests. Acceptance tests are those tests accomplished on equipment submitted under a contract to determine that the units are equivalent in performance and construction to the approved qualification samples.

*6.5 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is WRAMA/MMEBL, Robins Air Force Base, Georgia. Information pertaining to qualification of products may be obtained from WRAMA/MMEBL, Robins Air Force Base, Georgia 31098; Phone - Area Code: 912-926-6354.

6.6 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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